

Data Visualization and Storytelling

Veda Technologies – Data Analytics Internship Task 4 – COVID-19 Data Analysis using Power BI

1. Executive Summary

This report documents Task 4 of the Veda Technologies Data Analytics Internship: Data Visualization and Storytelling. A COVID-19 dataset was analyzed using Microsoft Power BI to create meaningful visualizations covering total cases, active cases, discharged cases, and deaths across Indian states and geographical zones. The objective was to turn structured data into a clear visual narrative rather than a collection of unrelated charts.

2. Task Objective

The task focuses on communicating trends, patterns, and insights from structured data. The analysis aims to compare COVID-19 indicators across geographical areas, understand the distribution of cases and deaths, examine active versus discharged cases, and present findings in a format that can be understood by both technical and non-technical stakeholders.

3. Tools and Technologies

Primary Tool: Microsoft Power BI

Dataset: COVID-19 dataset

Techniques: Data aggregation, categorical comparison, geographical analysis, data visualization and storytelling.

4. Analytical Scope

The analysis uses state and zone as geographical dimensions and examines measures related to total cases, active cases, discharged cases, and deaths. The visualizations are descriptive and show patterns present in the selected dataset. They should not be interpreted as causal explanations for differences between regions.

5. Visualization 1 – Count of Deaths by Zones

Question: How does the reported number of COVID-19 deaths vary across zones?

This visualization provides a high-level comparison of reported deaths among the geographical zones represented in the dataset. It makes regional differences easier to identify.

Takeaway: Reported COVID-19 deaths are distributed differently across zones, allowing comparatively higher and lower death counts to be identified.

6. Visualization 2 – Sum of Total Cases by Zone and State

Question: How are total reported COVID-19 cases distributed across zones and states?

This visualization provides a detailed geographical breakdown by combining zone and state, allowing individual states to be compared within their respective zones.

Takeaway: Total reported cases vary across states and zones, revealing differences in the geographical distribution of COVID-19 cases.

7. Visualization 3 – Sum of Active and Discharged Cases

Question: How do active cases compare with discharged cases?

This comparison provides two complementary views of case status. Active cases represent cases recorded as ongoing, while discharged cases represent cases that had moved out of the active category.

Takeaway: Comparing active and discharged cases provides a clearer view of the overall case status represented in the dataset.

8. Visualization 4 – Sum of Deaths by State

Question: How do reported COVID-19 deaths differ across individual states?

The state-level visualization provides a more granular view than the zone-level analysis and supports comparison between individual states.

Takeaway: Reported death totals differ between states, highlighting the value of state-level analysis alongside broader zone-level comparisons.

9. Visualization 5 – Count of Active Cases by Zone

Question: Which zones contain a higher concentration of reported active cases?

This visualization compares active cases across geographical zones and complements the total-case and death analyses by focusing specifically on ongoing cases.

Takeaway: Active COVID-19 cases are not distributed uniformly across zones, allowing areas with comparatively greater concentrations to be identified.

10. Data Storytelling Flow

The visual story progresses from broad regional context to detailed evidence: deaths by zone → total cases by zone and state → active versus discharged cases → deaths by state → active cases by zone. This ordering helps the viewer move from overall geographical comparison toward more detailed state and case-status analysis.

11. Key Observations

- COVID-19 indicators vary across geographical zones and states.
- State-level analysis provides more detail than zone-level aggregation alone.
- Active and discharged cases provide complementary views of case status.
- Reported death counts differ across states and zones.
- Multiple related visualizations provide a more complete story than a single metric.

12. Design Principles Applied

Each visual is connected to a specific analytical question. The report emphasizes comparison and interpretation rather than unnecessary decoration. Visuals are organized from context to evidence, and each visual is paired with a concise takeaway. The analysis avoids treating visual association as proof of causation.

13. Interview Questions

Dashboard vs data story: A dashboard is designed primarily for monitoring and interactive exploration, while a data story follows a deliberate narrative to explain a specific question or finding.

Choosing a chart: The chart should match the analytical purpose; bar charts support category comparisons, line charts support trends, and other chart types should be selected according to the relationship being communicated.

Explaining complex data: Start with the question, select relevant variables, arrange visuals logically, remove unnecessary complexity, and provide concise takeaways.

14. Deliverables

- Power BI report containing the COVID-19 analysis.
- Five analytical visualizations covering cases, active cases, discharged cases and deaths.
- A one-sentence takeaway for each visualization.
- Documentation describing the analytical narrative and findings.

15. Conclusion

This task demonstrates the practical application of data visualization and storytelling using Microsoft Power BI. The COVID-19 dataset was transformed into focused visualizations covering total cases, active cases, discharged cases, and deaths across Indian states and zones. The project reinforces the importance of choosing visuals based on analytical questions and presenting them as a coherent story rather than unrelated charts.

Internship: Veda Technologies – Data Analytics Internship

Task: Task 4 – Data Visualization and Storytelling

Tool: Microsoft Power BI

Dataset: COVID-19 Dataset